

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Product name	Rustilo HL 01
UFI:	CN10-S0VK-300G-DU3G
Product code	467096-DE04
SDS #	467096
Product type	Liquid.

1.2 Relevant identified uses of the substance or mixture and uses advised against

Identified uses
Use of lubricants and greases in open systems-Industrial Use of lubricants and greases in open systems-Professional

Use of the substance/ mixture	Corrosion inhibitor. For specific application advice see appropriate Technical Data Sheet or consult our company representative.
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1.3 Details of the supplier of the safety data sheet

Supplier	Castrol Holdings Europe B.V., d'Arcyweg 76, 3198NA Europoort Rotterdam Castrol Germany GmbH, Überseeallee 1, 20457 Hamburg +49 (0) 800 863 73 70 MSDSadvice@bp.com
E-mail address	

1.4 Emergency telephone number

EMERGENCY TELEPHONE NUMBER	Carechem: +44 (0) 1235 239 670 (24/7)
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SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Product definition	Mixture
Classification according to Regulation (EC) No. 1272/2008 [CLP/GHS]	

Skin Irrit. 2, H315
Eye Irrit. 2, H319
Aquatic Chronic 3, H412

See Section 16 for the full text of the H statements declared above.

See sections 11 and 12 for more detailed information on health effects and symptoms and environmental hazards.

2.2 Label elements

UFI:	CN10-S0VK-300G-DU3G
Hazard pictograms	



Signal word	Warning
Hazard statements	H315 - Causes skin irritation. H319 - Causes serious eye irritation. H412 - Harmful to aquatic life with long lasting effects.

Precautionary statements

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SECTION 2: Hazards identification

Prevention	P280 - Wear protective gloves. Wear eye or face protection. P273 - Avoid release to the environment. P264 - Wash hands thoroughly after handling.
Response	P302 + P352 - IF ON SKIN: Wash with plenty of soap and water. P362 + P364 - Take off contaminated clothing and wash it before reuse. P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. P337 + P313 - If eye irritation persists: Get medical attention.
Storage	Not applicable.
Disposal	P501 - Dispose of contents and container in accordance with all local, regional, national and international regulations.
Hazardous ingredients	Not applicable.
Supplemental label elements	Not applicable.
EU Regulation (EC) No. 1907/2006 (REACH)	
Annex XVII - Restrictions on the manufacture, placing on the market and use of certain dangerous substances, mixtures and articles	Not applicable.
Special packaging requirements	
Containers to be fitted with child-resistant fastenings	Not applicable.
Tactile warning of danger	Not applicable.
2.3 Other hazards	
Product meets the criteria for PBT or vPvB according to Regulation (EC) No. 1907/2006, Annex XIII	This mixture does not contain any substances that are assessed to be a PBT or a vPvB.
Product meets the criteria for endocrine disrupting properties according to Regulation (EC) No. 1907/2006.	This substance/mixture does not contain any components that are considered to have endocrine disrupting properties.
Other hazards which do not result in classification	Defatting to the skin.

SECTION 3: Composition/information on ingredients

3.2 Mixtures

Product definition Mixture
Highly refined base oil (IP 346 DMSO extract < 3%). Proprietary performance additives.

Product/ingredient name	Identifiers	%	Classification	Specific Conc. Limits, M-factors and ATEs	Type
Distillates (petroleum), solvent-dewaxed light paraffinic	REACH #: 01-2119480132-48 EC: 265-159-2 CAS: 64742-56-9 Index: 649-469-00-9	≤10	Asp. Tox. 1, H304	-	[1]
dicyclohexylamine	REACH #: 01-2119493354-33 EC: 202-980-7 CAS: 101-83-7 Index: 612-066-00-3	≤1.9	Acute Tox. 3, H301 Acute Tox. 3, H311 Skin Corr. 1B, H314 Eye Dam. 1, H318 Aquatic Acute 1, H400 Aquatic Chronic 1, H410	ATE [Oral] = 100 mg/kg ATE [Dermal] = 300 mg/kg M [Acute] = 1 M [Chronic] = 1	[1] [2]
2,6-ditert-butyl-p-cresol	REACH #: 01-2119555270-46 EC: 204-881-4 CAS: 128-37-0	≤0.59	Aquatic Acute 1, H400 Aquatic Chronic 1, H410	M [Acute] = 1 M [Chronic] = 1	[1] [2]

See Section 16 for the full text of the H statements declared above.

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SECTION 3: Composition/information on ingredients**Type**

[1] Substance classified with a health or environmental hazard

[2] Substance with a workplace exposure limit

Occupational exposure limits, if available, are listed in Section 8.

SECTION 4: First aid measures**4.1 Description of first aid measures****Eye contact**

In case of contact, immediately flush eyes with plenty of water for at least 15 minutes. Eyelids should be held away from the eyeball to ensure thorough rinsing. Check for and remove any contact lenses. Get medical attention.

Skin contact

Wash skin thoroughly with soap and water or use recognised skin cleanser. Remove contaminated clothing and shoes. Wash clothing before reuse. Clean shoes thoroughly before reuse. Get medical attention.

Inhalation

If inhaled, remove to fresh air. In case of inhalation of decomposition products in a fire, symptoms may be delayed. The exposed person may need to be kept under medical surveillance for 48 hours. Get medical attention if symptoms occur.

Ingestion

Do not induce vomiting unless directed to do so by medical personnel. Never give anything by mouth to an unconscious person. If unconscious, place in recovery position and get medical attention immediately. Get medical attention if symptoms occur.

Protection of first-aiders

No action shall be taken involving any personal risk or without suitable training. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation.

4.2 Most important symptoms and effects, both acute and delayed

See Section 11 for more detailed information on health effects and symptoms.

Potential acute health effects**Inhalation**

Exposure to decomposition products may cause a health hazard. Serious effects may be delayed following exposure.

Ingestion

Irritating to mouth, throat and stomach.

Skin contact

Causes skin irritation. Defatting to the skin.

Eye contact

See: Section 11. Toxicological Information - Potential acute health effects: Eye contact

Delayed and immediate effects as well as chronic effects from short and long-term exposure**Inhalation**

Overexposure to the inhalation of airborne droplets or aerosols may cause irritation of the respiratory tract.

Ingestion

Ingestion of large quantities may cause nausea and diarrhoea.

Skin contact

Prolonged or repeated contact can defat the skin and lead to irritation and/or dermatitis.

Eye contact

Potential risk of transient stinging or redness if accidental eye contact occurs.

4.3 Indication of any immediate medical attention and special treatment needed**Notes to physician**

Treatment should in general be symptomatic and directed to relieving any effects. In case of inhalation of decomposition products in a fire, symptoms may be delayed. The exposed person may need to be kept under medical surveillance for 48 hours.

SECTION 5: Firefighting measures**5.1 Extinguishing media****Suitable extinguishing media**

Use foam or all-purpose dry chemical to extinguish.

Unsuitable extinguishing media

Do not use water jet. The use of a water jet may cause the fire to spread by splashing the burning product.

5.2 Special hazards arising from the substance or mixture**Hazards from the substance or mixture**

In a fire or if heated, a pressure increase will occur and the container may burst.

Hazardous combustion products

Combustion products may include the following:
carbon oxides (CO, CO₂) (carbon monoxide, carbon dioxide)
nitrogen oxides (NO, NO₂ etc.)

5.3 Advice for firefighters**Special precautions for fire-fighters**

No action shall be taken involving any personal risk or without suitable training. Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. This material is harmful to aquatic organisms. Fire water contaminated with this material must be contained and prevented from being discharged to any waterway, sewer or drain.

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SECTION 5: Firefighting measures

Special protective equipment for fire-fighters

Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode. Clothing for fire-fighters (including helmets, protective boots and gloves) conforming to European standard EN 469 will provide a basic level of protection for chemical incidents.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

For non-emergency personnel

Contact emergency personnel. No action shall be taken involving any personal risk or without suitable training. Evacuate surrounding areas. Keep unnecessary and unprotected personnel from entering. Do not touch or walk through spilt material. Floors may be slippery; use care to avoid falling. Avoid breathing vapour or mist. Provide adequate ventilation. Put on appropriate personal protective equipment.

For emergency responders

Entry into a confined space or poorly ventilated area contaminated with vapour, mist or fume is extremely hazardous without the correct respiratory protective equipment and a safe system of work. Wear self-contained breathing apparatus. Wear a suitable chemical protective suit. Chemical resistant boots. See also the information in "For non-emergency personnel".

6.2 Environmental precautions

Avoid dispersal of spilt material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air). Water polluting material. May be harmful to the environment if released in large quantities.

6.3 Methods and material for containment and cleaning up

Small spill

Stop leak if without risk. Move containers from spill area. Absorb with an inert material and place in an appropriate waste disposal container. Dispose of via a licensed waste disposal contractor.

Large spill

Stop leak if without risk. Move containers from spill area. Approach the release from upwind. Prevent entry into sewers, water courses, basements or confined areas. Contain and collect spillage with non-combustible, absorbent material e.g. sand, earth, vermiculite or diatomaceous earth and place in container for disposal according to local regulations. Contaminated absorbent material may pose the same hazard as the spilt product. Dispose of via a licensed waste disposal contractor.

6.4 Reference to other sections

See Section 1 for emergency contact information.
See Section 5 for firefighting measures.
See Section 8 for information on appropriate personal protective equipment.
See Section 12 for environmental precautions.
See Section 13 for additional waste treatment information.

SECTION 7: Handling and storage

The information in this section contains generic advice and guidance. The list of Identified Uses in Section 1 should be consulted for any available use-specific information provided in the Exposure Scenario(s).

7.1 Precautions for safe handling

Protective measures

Put on appropriate personal protective equipment. Do not ingest. Avoid contact with eyes, skin and clothing. Avoid breathing vapour or mist. Avoid contact of spilt material and runoff with soil and surface waterways. Keep in the original container or an approved alternative made from a compatible material, kept tightly closed when not in use. Do not reuse container. Empty containers retain product residue and can be hazardous.

Advice on general occupational hygiene

Eating, drinking and smoking should be prohibited in areas where this material is handled, stored and processed. Wash thoroughly after handling. Remove contaminated clothing and protective equipment before entering eating areas. See also Section 8 for additional information on hygiene measures.

7.2 Conditions for safe storage, including any incompatibilities

Store in accordance with local regulations. Store in a dry, cool and well-ventilated area, away from incompatible materials (see Section 10). Keep away from heat and direct sunlight. Keep container tightly closed and sealed until ready for use. Containers that have been opened must be carefully resealed and kept upright to prevent leakage. Store and use only in equipment/containers designed for use with this product. Do not store in unlabelled containers. Use appropriate containment to avoid environmental contamination.

Not suitable

Prolonged exposure to elevated temperature

Germany - Storage code

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7.3 Specific end use(s)

Recommendations

See section 1.2 and Exposure scenarios in annex, if applicable.

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SECTION 8: Exposure controls/personal protection

The information in this section contains generic advice and guidance. The list of Identified Uses in Section 1 should be consulted for any available use-specific information provided in the Exposure Scenario(s).

8.1 Control parameters

Occupational exposure limits

Product/ingredient name	Exposure limit values
dicyclohexylamine	TRGS 900 OEL (Germany) Absorbed through skin. TWA 8 hours: 5 mg/m³. Issued/Revised: 7/2013. TWA 8 hours: 0.7 ppm. Issued/Revised: 7/2013. PEAK 15 minutes: 10 mg/m³. Issued/Revised: 7/2013. PEAK 15 minutes: 1.4 ppm. Issued/Revised: 7/2013.
2,6-ditert-butyl-p-cresol	TRGS 900 OEL (Germany) TWA 8 hours: 10 mg/m³. Form: Inhalable fraction. Issued/Revised: 7/2013. PEAK 15 minutes: 40 mg/m³. Form: Inhalable fraction. Issued/Revised: 7/2013.

Whilst specific OELs for certain components may be shown in this section, other components may be present in any mist, vapour or dust produced. Therefore, the specific OELs may not be applicable to the product as a whole and are provided for guidance only.

Recommended monitoring procedures	Reference should be made to monitoring standards, such as the following: European Standard EN 689 (Workplace atmospheres - Guidance for the assessment of exposure by inhalation to chemical agents for comparison with limit values and measurement strategy) European Standard EN 14042 (Workplace atmospheres - Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents) European Standard EN 482 (Workplace atmospheres - General requirements for the performance of procedures for the measurement of chemical agents) Reference to national guidance documents for methods for the determination of hazardous substances will also be required.
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Biological exposure indices

Product/ingredient name	Exposure indices
2,6-ditert-butyl-p-cresol	DFG BEI-values list (Germany, 7/2024) BEI: 7 µg/l, Butylated hydroxytoluene acid (after hydrolysis) [in urine]. Sampling time: end of exposure or end of shift.

DNELs/DMELs

Product/ingredient name	Result
dicyclohexylamine	DNEL - Workers - Long term - Inhalation 0.353 mg/m³ Effects: Systemic DNEL - Workers - Long term - Dermal 0.1 mg/kg bw/day Effects: Systemic
2,6-di-tert-butyl-p-cresol	DNEL - Workers - Long term - Inhalation 1.76 mg/m³ Effects: Systemic DNEL - Workers - Long term - Dermal 0.5 mg/kg bw/day Effects: Systemic DNEL - General population - Long term - Inhalation 0.435 mg/m³ Effects: Systemic DNEL - General population - Long term - Dermal 0.25 mg/kg bw/day Effects: Systemic DNEL - General population - Long term - Oral 0.25 mg/kg bw/day Effects: Systemic

PNECs

Product/ingredient name	Result
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SECTION 8: Exposure controls/personal protection

dicyclohexylamine

Fresh water

0.002 mg/l

Marine water

0 mg/l

Sewage Treatment Plant

21 mg/l

Fresh water sediment

0.075 mg/kg dwt

Marine water sediment

0.007 mg/kg dwt

Soil

0.014 mg/kg dwt

2,6-di-tert-butyl-p-cresol

Fresh water

0.199 µg/l

Marine water

0.02 µg/l

Sewage Treatment Plant

0.17 mg/l

Fresh water sediment

0.458 mg/l

Marine water sediment

0.046 mg/l

Soil

0.054 mg/l

8.2 Exposure controls**Appropriate engineering controls**

Provide exhaust ventilation or other engineering controls to keep the relevant airborne concentrations below their respective occupational exposure limits.

All activities involving chemicals should be assessed for their risks to health, to ensure exposures are adequately controlled. Personal protective equipment should only be considered after other forms of control measures (e.g. engineering controls) have been suitably evaluated. Personal protective equipment should conform to appropriate standards, be suitable for use, be kept in good condition and properly maintained.

Your supplier of personal protective equipment should be consulted for advice on selection and appropriate standards. For further information contact your national organisation for standards. The final choice of protective equipment will depend upon a risk assessment. It is important to ensure that all items of personal protective equipment are compatible.

Individual protection measures**Hygiene measures**

Wash hands, forearms and face thoroughly after handling chemical products, before eating, smoking and using the lavatory and at the end of the working period. Ensure that eyewash stations and safety showers are close to the workstation location.

Respiratory protection

In case of insufficient ventilation, wear suitable respiratory equipment.

For protection against metal working fluids, respiratory protection that is classified as "resistant to oil" (class R) or oil proof (class P) should be selected where appropriate. Depending on the level of airborne contaminants, an air-purifying, half-mask respirator (with HEPA filter) including disposable (P- or R-series) (for oil mists less than 50mg/m³), or any powered, air-purifying respirator equipped with hood or helmet and HEPA filter (for oil mists less than 125 mg/m³). Where organic vapours are a potential hazard during metalworking operations, a combination particulate and organic vapour filter may be necessary.

The correct choice of respiratory protection depends upon the chemicals being handled, the conditions of work and use, and the condition of the respiratory equipment. Safety procedures should be developed for each intended application. Respiratory protection equipment should therefore be chosen in consultation with the supplier/manufacturer and with a full assessment of the working conditions.

Eye/face protection

Safety glasses with side shields.

Skin protection**Product name** Rustilo HL 01**Product code** 467096-DE04**Page:** 6/23**Version** 15 **Date of issue** 8 January 2026**Format** Germany**Language** ENGLISH**Date of previous issue** 6 September 2023.**(Germany)**

SECTION 8: Exposure controls/personal protection

Hand protection

General Information:

Because specific work environments and material handling practices vary, safety procedures should be developed for each intended application. The correct choice of protective gloves depends upon the chemicals being handled, and the conditions of work and use. Most gloves provide protection for only a limited time before they must be discarded and replaced (even the best chemically resistant gloves will break down after repeated chemical exposures).

Gloves should be chosen in consultation with the supplier / manufacturer and taking account of a full assessment of the working conditions.

Wear suitable gloves.

Recommended: Nitrile gloves.

Breakthrough time:

Breakthrough time data are generated by glove manufacturers under laboratory test conditions and represent how long a glove can be expected to provide effective permeation resistance. It is important when following breakthrough time recommendations that actual workplace conditions are taken into account. Always consult with your glove supplier for up-to-date technical information on breakthrough times for the recommended glove type.

Our recommendations on the selection of gloves are as follows:

Continuous contact:

Gloves with a minimum breakthrough time of 240 minutes, or >480 minutes if suitable gloves can be obtained.

If suitable gloves are not available to offer that level of protection, gloves with shorter breakthrough times may be acceptable as long as appropriate glove maintenance and replacement regimes are determined and adhered to.

Short-term / splash protection:

Recommended breakthrough times as above.

It is recognised that for short-term, transient exposures, gloves with shorter breakthrough times may commonly be used. Therefore, appropriate maintenance and replacement regimes must be determined and rigorously followed.

Glove Thickness:

For general applications, we recommend gloves with a thickness typically greater than 0.35 mm.

It should be emphasised that glove thickness is not necessarily a good predictor of glove resistance to a specific chemical, as the permeation efficiency of the glove will be dependent on the exact composition of the glove material. Therefore, glove selection should also be based on consideration of the task requirements and knowledge of breakthrough times.

Glove thickness may also vary depending on the glove manufacturer, the glove type and the glove model. Therefore, the manufacturers' technical data should always be taken into account to ensure selection of the most appropriate glove for the task.

Note: Depending on the activity being conducted, gloves of varying thickness may be required for specific tasks. For example:

- Thinner gloves (down to 0.1 mm or less) may be required where a high degree of manual dexterity is needed. However, these gloves are only likely to give short duration protection and would normally be just for single use applications, then disposed of.

- Thicker gloves (up to 3 mm or more) may be required where there is a mechanical (as well as a chemical) risk i.e. where there is abrasion or puncture potential.

Skin and body

Use of protective clothing is good industrial practice.

Personal protective equipment for the body should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product.

Cotton or polyester/cotton overalls will only provide protection against light superficial contamination that will not soak through to the skin. Overalls should be laundered on a regular basis. When the risk of skin exposure is high (e.g. when cleaning up spillages or if there is a risk of splashing) then chemical resistant aprons and/or impervious chemical suits and boots will be required.

SECTION 8: Exposure controls/personal protection

Refer to standards:	Respiratory protection: EN 529 Gloves: EN 420, EN 374 Eye protection: EN 166 Filtering half-mask: EN 149 Filtering half-mask with valve: EN 405 Half-mask: EN 140 plus filter Full-face mask: EN 136 plus filter Particulate filters: EN 143 Gas/combined filters: EN 14387
Environmental exposure controls	Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

SECTION 9: Physical and chemical properties

The conditions of measurement of all properties are at standard temperature and pressure unless otherwise indicated.

9.1 Information on basic physical and chemical properties

Physical state	Liquid.																																	
Colour	Yellow. [Light]																																	
Odour	Not available.																																	
Odour threshold	Not available.																																	
Melting point/freezing point	Not available.																																	
Initial boiling point and boiling range	Not available.																																	
Flammability	Not available.																																	
Lower and upper explosion limit	Not available.																																	
Flash point	Open cup: >191°C (>375.8°F) [Cleveland DIN EN ISO 2592]																																	
Auto-ignition temperature	<table><tr><th>Ingredient name</th><th>°C</th><th>°F</th><th>Method</th></tr><tr><td>dicyclohexylamine</td><td>255</td><td>491</td><td></td></tr></table>	Ingredient name	°C	°F	Method	dicyclohexylamine	255	491																										
Ingredient name	°C	°F	Method																															
dicyclohexylamine	255	491																																
Decomposition temperature	Not available.																																	
pH	Not applicable.																																	
Kinematic viscosity	Kinematic: 27 mm²/s (27 cSt) at 40°C Kinematic: 71.1 mm²/s (71.1 cSt) at 20°C (DIN EN ISO 3104)																																	
Solubility	<table><tr><th>Media</th><th>Result</th></tr><tr><td>water</td><td>Not soluble</td></tr></table>	Media	Result	water	Not soluble																													
Media	Result																																	
water	Not soluble																																	
Partition coefficient n-octanol/ water (log value)	Not applicable.																																	
Vapour pressure	<table><tr><th rowspan="2">Ingredient name</th><th colspan="2">Vapour Pressure at 20°C</th><th colspan="3">Vapour pressure at 50°C</th></tr><tr><th>mm Hg</th><th>kPa</th><th>Method</th><th>mm Hg</th><th>kPa</th><th>Method</th></tr><tr><td>Distillates (petroleum), hydrotreated heavy paraffinic</td><td><0.07501</td><td><0.01</td><td>ASTM D 5191</td><td></td><td></td><td></td></tr><tr><td>Distillates (petroleum), solvent-dewaxed light paraffinic</td><td><0.07501</td><td><0.01</td><td>ASTM D 5191</td><td></td><td></td><td></td></tr><tr><td>dicyclohexylamine</td><td>0.056</td><td>0.0075</td><td>EU A.4</td><td></td><td></td><td></td></tr></table>	Ingredient name	Vapour Pressure at 20°C		Vapour pressure at 50°C			mm Hg	kPa	Method	mm Hg	kPa	Method	Distillates (petroleum), hydrotreated heavy paraffinic	<0.07501	<0.01	ASTM D 5191				Distillates (petroleum), solvent-dewaxed light paraffinic	<0.07501	<0.01	ASTM D 5191				dicyclohexylamine	0.056	0.0075	EU A.4			
Ingredient name	Vapour Pressure at 20°C		Vapour pressure at 50°C																															
	mm Hg	kPa	Method	mm Hg	kPa	Method																												
Distillates (petroleum), hydrotreated heavy paraffinic	<0.07501	<0.01	ASTM D 5191																															
Distillates (petroleum), solvent-dewaxed light paraffinic	<0.07501	<0.01	ASTM D 5191																															
dicyclohexylamine	0.056	0.0075	EU A.4																															
Density and/or Relative density	<1000 kg/m³ (<1 g/cm³) at 15°C																																	
Relative vapour density	Not available.																																	
Particle characteristics																																		
Median particle size	Not applicable.																																	
9.2 Other information																																		
Evaporation rate	Not available.																																	
Explosive properties	Not available.																																	
Oxidising properties	Not available.																																	

SECTION 9: Physical and chemical properties

Pour point <-22 °C

SECTION 10: Stability and reactivity

10.1 Reactivity	No specific test data available for this product. Refer to Conditions to avoid and Incompatible materials for additional information.
10.2 Chemical stability	The product is stable.
10.3 Possibility of hazardous reactions	Under normal conditions of storage and use, hazardous reactions will not occur. Under normal conditions of storage and use, hazardous polymerisation will not occur.
10.4 Conditions to avoid	Avoid all possible sources of ignition (spark or flame).
10.5 Incompatible materials	Reactive or incompatible with the following materials: oxidising materials.
10.6 Hazardous decomposition products	Under normal conditions of storage and use, hazardous decomposition products should not be produced.

SECTION 11: Toxicological information

11.1 Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute toxicity

Product/ingredient name	Result
Distillates (petroleum), solvent-dewaxed light paraffinic	Rat - Oral - LD50 >5000 mg/kg OECD 401
	Rabbit - Dermal - LD50 >5000 mg/kg OECD 402
	Rat - Inhalation - LC50 Dusts and mists >2.18 mg/l [4 hours] OECD 403
dicyclohexylamine	Rat - Oral - LD50 200 mg/kg
	Rabbit - Dermal - LD50 200 to 316 mg/kg
	Rat - Inhalation - LC50 Vapour >1.4 mg/l [6 hours]
2,6-di-tert-butyl-p-cresol	Rat - Oral - LD50 >5000 mg/kg OECD 401
	Rat - Dermal - LD50 >2000 mg/kg OECD 402

Acute toxicity estimates

Product/ingredient name	Oral (mg/kg)	Dermal (mg/kg)	Inhalation (gases) (ppm)	Inhalation (vapours) (mg/l)	Inhalation (dusts and mists) (mg/l)
Rustilo HL 01 dicyclohexylamine	6251.9 100	18755.6 300	N/A N/A	N/A N/A	N/A N/A

Skin corrosion/irritation

Product/ingredient name	Result
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SECTION 11: Toxicological information

Distillates (petroleum), solvent-dewaxed light paraffinic	Rabbit - Skin - Non-irritant to skin.
dicyclohexylamine	Rabbit - Skin - Corrosive
2,6-di-tert-butyl-p-cresol	Rabbit - Skin - Non-irritant to skin. OECD 404
Serious eye damage/eye irritation	
Product/ingredient name	
Distillates (petroleum), solvent-dewaxed light paraffinic	Result Rabbit - Eyes - Non-irritating to the eyes. OECD 405
dicyclohexylamine	Rabbit - Eyes - Severe irritant
2,6-di-tert-butyl-p-cresol	Rabbit - Eyes - Non-irritating to the eyes. OECD 405
Respiratory corrosion/irritation	
Not available.	
Respiratory or skin sensitization	
Product/ingredient name	
Distillates (petroleum), solvent-dewaxed light paraffinic	Result Guinea pig - skin OECD 406 Result: Not sensitising
2,6-di-tert-butyl-p-cresol	Guinea pig - skin OECD 406 Result: Not sensitising
Germ cell mutagenicity	
Product/ingredient name	
Distillates (petroleum), solvent-dewaxed light paraffinic	Result In vitro - Bacteria OECD [Bacterial Reverse Mutation Test] Result: Negative In vitro - Mammal - species unspecified In vitro Mammalian Chromosomal Aberration Test Result: Negative
dicyclohexylamine	In vitro - Bacteria Bacterial Reverse Mutation Test Result: Negative In vitro - Mammal - species unspecified Result: Negative In vitro - Mammal - species unspecified Mammalian Erythrocyte Micronucleus Test Result: Negative In vivo - Mammal - species unspecified Genetic Toxicology: Rodent Dominant Lethal Test Result: Negative
2,6-di-tert-butyl-p-cresol	In vitro - Bacteria Bacterial Reverse Mutation Test Result: Negative In vivo - Mammal - species unspecified Mammalian Erythrocyte Micronucleus Test Result: Negative

SECTION 11: Toxicological information**Carcinogenicity****Product/ingredient name**

2,6-di-tert-butyl-p-cresol

Result**Rat - Oral - Unspecified**Result: Negative**Reproductive toxicity****Product/ingredient name**

Distillates (petroleum), solvent-dewaxed light paraffinic

Result**Rat - Oral**

OECD 421

Maternal toxicity: NegativeFertility effects: NegativeDevelopmental: Negative

dicyclohexylamine

Rat - Oral

OECD 421

Maternal toxicity: PositiveFertility effects: NegativeDevelopmental: Negative

2,6-di-tert-butyl-p-cresol

Rat - Oral

OECD 416

Maternal toxicity: PositiveFertility effects: NegativeDevelopmental: Negative**Specific target organ toxicity (single exposure)**

Not available.

Specific target organ toxicity (repeated exposure)

Not available.

Aspiration hazard**Product/ingredient name**

Distillates (petroleum), solvent-dewaxed light paraffinic

Result

ASPIRATION HAZARD - Category 1

Information on likely routes of exposure

Routes of entry anticipated: Oral, Dermal, Inhalation, Eyes.

Potential acute health effects**Inhalation**

Exposure to decomposition products may cause a health hazard. Serious effects may be delayed following exposure.

Ingestion

Irritating to mouth, throat and stomach.

Skin contact

Causes skin irritation. Defatting to the skin.

Eye contact

Causes serious eye irritation.

Symptoms related to the physical, chemical and toxicological characteristics**Inhalation**

No specific data.

Ingestion

No specific data.

Skin contactAdverse symptoms may include the following:
irritation
redness
dryness
cracking**Eye contact**Adverse symptoms may include the following:
pain or irritation
watering
redness**Delayed and immediate effects as well as chronic effects from short and long-term exposure****Inhalation**

Overexposure to the inhalation of airborne droplets or aerosols may cause irritation of the respiratory tract.

Ingestion

Ingestion of large quantities may cause nausea and diarrhoea.

Product name Rustilo HL 01**Product code** 467096-DE04**Page:** 11/23**Version** 15 **Date of issue** 8 January 2026**Format** Germany**Language** ENGLISH**Date of previous issue** 6 September 2023.**(Germany)**

SECTION 11: Toxicological information

Skin contact	Prolonged or repeated contact can defat the skin and lead to irritation and/or dermatitis.
Eye contact	Potential risk of transient stinging or redness if accidental eye contact occurs.
Potential chronic health effects	
Not available.	
Conclusion/Summary [Product]	Not available.
General	No known significant effects or critical hazards.
Carcinogenicity	No known significant effects or critical hazards.
Mutagenicity	No known significant effects or critical hazards.
Developmental effects	No known significant effects or critical hazards.
Fertility effects	No known significant effects or critical hazards.

11.2 Information on other hazards

11.2.1 Endocrine disrupting properties

Conclusion/Summary [Product]	This substance/mixture does not contain any components that are considered to have endocrine disrupting properties.
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11.2.2 Other information

Not available.

SECTION 12: Ecological information

12.1 Toxicity

Product/ingredient name	Result
Distillates (petroleum), solvent-dewaxed light paraffinic	Acute - EL50 OECD 201 Algae >100 mg/l [72 hours]
	Acute - EL50 OECD 202 Daphnia >10000 mg/l [48 hours]
	Acute - LL50 OECD 203 Fish >100 mg/l [96 hours]
	Chronic - NOEL OECD 201 Algae ≥100 mg/l [72 hours]
	Chronic - NOEL OECD 211 Daphnia 10 mg/l [21 days]
	Acute - ErC50 OECD 201 Algae 1 mg/l [72 hours]
	Acute - EC50 DIN 38412 Part 11 Daphnia 43 mg/l [48 hours]
	Acute - LC50 OECD 203 Fish 62 mg/l [96 hours]
	Acute - EC50 DIN 38412 Part 8 Micro-organism 201 mg/l [17 hours]
	Chronic - NOEC

dicyclohexylamine

SECTION 12: Ecological information

2,6-di-tert-butyl-p-cresol	OECD 201 Daphnia 2 mg/l [72 hours]
	Chronic - NOEC OECD 211 Fish 0.016 mg/l [21 days]
	Acute - EC50 OECD 201 Algae >0.4 mg/l [72 hours]
	Acute - EC50 OECD 202 Daphnia 0.48 mg/l [48 hours]
	Acute - LC50 OECD 203 Fish >0.57 mg/l [96 hours]
	Chronic - NOEC OECD 201 Algae 0.4 mg/l [72 hours]
	Chronic - NOEC OECD 211 Daphnia 0.069 mg/l [21 days]
	Chronic - NOEC OECD 210 Fish 0.053 mg/l [30 days]

Environmental hazards Harmful to aquatic life with long lasting effects.

12.2 Persistence and degradability

Not expected to be rapidly degradable.

Product/ingredient name	Result
Distillates (petroleum), solvent-dewaxed light paraffinic	OECD 301F 31% [28 days] - Not readily
dicyclohexylamine	OECD 301D 96% [20 days] - Readily
2,6-di-tert-butyl-p-cresol	OECD 301C 4.5% [28 days] - Not readily

Product/ingredient name	Aquatic half-life	Photolysis	Biodegradability
dicyclohexylamine	-	-	Readily

12.3 Bioaccumulative potential

Not available.

Product/ingredient name	LogP _{ow}	BCF	Potential
dicyclohexylamine	2.724	-	Low
2,6-di-tert-butyl-p-cresol	5.1	330 to 1800 [OECD 305 C]	High

12.4 Mobility in soil

Soil/water partition coefficient

Product/ingredient name	logK _{oc}	K _{oc}
dicyclohexylamine	2.5	347.622
2,6-di-tert-butyl-p-cresol	3.7	4489.84

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SECTION 12: Ecological information

Results of PMT and vPvM assessment

Product/ingredient name	PMT	P	M	T	vPvM	vP	vM
Distillates (petroleum), solvent-dewaxed light paraffinic	No	No	No	No	No	No	No
dicyclohexylamine	No	No	No	No	No	No	No
2,6-di-tert-butyl-p-cresol	No	No	No	No	No	No	No

Mobility Liquid. insoluble in water.

Conclusion/Summary The product does not meet the criteria to be considered as a PMT or vPvM.

12.5 Results of PBT and vPvB assessment

Regulation (EC) No. 1907/2006 [REACH]

Product/ingredient name	PBT	P	B	T	vPvB	vP	vB
Distillates (petroleum), solvent-dewaxed light paraffinic	No	N/A	N/A	No	N/A	N/A	N/A
dicyclohexylamine	No	N/A	N/A	No	N/A	N/A	N/A
2,6-di-tert-butyl-p-cresol	No	N/A	No	No	No	N/A	No

Regulation (EC) No. 1272/2008 [CLP]

Product/ingredient name	PBT	P	B	T	vPvB	vP	vB
Distillates (petroleum), solvent-dewaxed light paraffinic	No	No	No	No	No	No	No
dicyclohexylamine	No	No	No	No	No	No	No
2,6-di-tert-butyl-p-cresol	No	No	No	No	No	No	No

Conclusion/Summary The product does not meet the criteria to be considered as a PBT or vPvB.

Regulation (EC) No. 1272/2008 [CLP]

12.6 Endocrine disrupting properties

Conclusion/Summary [Product] This substance/mixture does not contain any components that are considered to have endocrine disrupting properties.

12.7 Other adverse effects No known significant effects or critical hazards.

SECTION 13: Disposal considerations

The information in this section contains generic advice and guidance. The list of Identified Uses in Section 1 should be consulted for any available use-specific information provided in the Exposure Scenario(s).

13.1 Waste treatment methods

Product

Methods of disposal Where possible, arrange for product to be recycled. Dispose of via an authorised person/ licensed waste disposal contractor in accordance with local regulations.

Hazardous waste Yes.

European waste catalogue (EWC)

Waste code	Waste designation
12 01 07*	mineral-based machining oils free of halogens (except emulsions and solutions)

However, deviation from the intended use and/or the presence of any potential contaminants may require an alternative waste disposal code to be assigned by the end user.

Packaging

Methods of disposal Where possible, arrange for product to be recycled. Dispose of via an authorised person/ licensed waste disposal contractor in accordance with local regulations.

Waste code	European waste catalogue (EWC)
15 01 10*	packaging containing residues of or contaminated by hazardous substances

Special precautions This material and its container must be disposed of in a safe way. Care should be taken when handling emptied containers that have not been cleaned or rinsed out. Empty containers or liners may retain some product residues. Empty containers represent a fire hazard as they may contain flammable product residues and vapour. Never weld, solder or braze empty containers. Avoid dispersal of spill material and runoff and contact with soil, waterways, drains and sewers.

SECTION 13: Disposal considerations**References**Commission 2014/955/EU
Directive 2008/98/EC**SECTION 14: Transport information**

	ADR/RID	ADN	IMDG	IATA
14.1 UN number or ID number	Not regulated.	Not regulated.	Not regulated.	Not regulated.
14.2 UN proper shipping name	-	-	-	-
14.3 Transport hazard class(es)	-	-	-	-
14.4 Packing group	-	-	-	-
14.5 Environmental hazards	No.	No.	No.	No.
Additional information	-	-	-	-

14.6 Special precautions for user Not available.**14.7 Maritime transport in bulk according to IMO instruments** Not available.**SECTION 15: Regulatory information****15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture**[EU Regulation \(EC\) No. 1907/2006 \(REACH\)](#)[Annex XIV - List of substances subject to authorisation](#)[Annex XIV](#)

None of the components are listed.

[Substances of very high concern](#)

None of the components are listed.

[Annex XVII - Restrictions on the manufacture, placing on the market and use of certain dangerous substances, mixtures and articles](#)

Product/ingredient name	%	Designation [Usage]
Rustilo HL 01 methanol	95-100 <0.01	3 69

Labelling Not applicable.**Other regulations****REACH Status**

The company, as identified in Section 1, sells this product in the EU in compliance with the current requirements of REACH.

United States inventory (TSCA 8b)

All components are active or exempted.

Australia inventory (AIC)

All components are listed or exempted.

Canada inventory

All components are listed or exempted.

China inventory (IECSC)

All components are listed or exempted.

Japan inventory (CSCL)

All components are listed or exempted.

Korea inventory (KECI)

All components are listed or exempted.

Philippines inventory (PICCS)

All components are listed or exempted.

Taiwan Chemical Substances Inventory (TCSI)

All components are listed or exempted.

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SECTION 15: Regulatory information

Explosive precursors	Not applicable.
Ozone depleting substances (EU 2024/590)	Not listed.
Prior Informed Consent (PIC) (649/2012/EU)	Not listed.
Persistent Organic Pollutants	Not listed.
EU - Water framework directive - Priority substances	None of the components are listed.
Seveso Directive	This product is not controlled under the Seveso Directive.
National regulations	
Hazardous incident ordinance	This product is not controlled under the Germany Hazardous Incident Ordinance.
Hazard class for water	1 (classified according AwsV)
Prohibited Chemicals Regulation (ChemVerbotsV)	When placed on the market in Germany, this product is not subject to the Prohibited Chemicals Regulation (ChemVerbotsV).
Occupational restrictions	Observe employment restrictions in the following: Gesetz zum Schutz der arbeitenden Jugend (Jugendarbeitsschutzgesetz – JArbSchG) Gesetz zum Schutz von Müttern bei der Arbeit, in der Ausbildung und im Studium (Mutterschutzgesetz – MuSchG)
15.2 Chemical safety assessment	A Chemical Safety Assessment has been carried out for one or more of the substances within this mixture. A Chemical Safety Assessment has not been carried out for the mixture itself.

SECTION 16: Other information

Abbreviations and acronyms	ADN = European Provisions concerning the International Carriage of Dangerous Goods by Inland Waterway ADR = The European Agreement concerning the International Carriage of Dangerous Goods by Road ATE = Acute Toxicity Estimate BCF = Bioconcentration Factor CAS = Chemical Abstracts Service CLP = Classification, Labelling and Packaging Regulation [Regulation (EC) No. 1272/2008] CSA = Chemical Safety Assessment CSR = Chemical Safety Report DMEL = Derived Minimal Effect Level DNEL = Derived No Effect Level EINECS = European Inventory of Existing Commercial chemical Substances ES = Exposure Scenario EUH statement = CLP-specific Hazard statement EWC = European Waste Catalogue GHS = Globally Harmonized System of Classification and Labelling of Chemicals IATA = International Air Transport Association IBC = Intermediate Bulk Container IMDG = International Maritime Dangerous Goods LogPow = logarithm of the octanol/water partition coefficient MARPOL = International Convention for the Prevention of Pollution From Ships, 1973 as modified by the Protocol of 1978. ("Marpol" = marine pollution) OECD = Organisation for Economic Co-operation and Development PBT = Persistent, Bioaccumulative and Toxic PNEC = Predicted No Effect Concentration REACH = Registration, Evaluation, Authorisation and Restriction of Chemicals Regulation [Regulation (EC) No. 1907/2006] RID = The Regulations concerning the International Carriage of Dangerous Goods by Rail RRN = REACH Registration Number SADT = Self-Accelerating Decomposition Temperature SVHC = Substances of Very High Concern STOT-RE = Specific Target Organ Toxicity - Repeated Exposure STOT-SE = Specific Target Organ Toxicity - Single Exposure TWA = Time weighted average UN = United Nations
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SECTION 16: Other information

UVCB = Complex hydrocarbon substance
 VOC = Volatile Organic Compound
 vPvB = Very Persistent and Very Bioaccumulative
 Varies = may contain one or more of the following 64741-88-4 / RRN 01-2119488706-23,
 64741-89-5 / RRN 01-2119487067-30, 64741-95-3 / RRN 01-2119487081-40, 64741-96-4 / RRN
 01-2119483621-38, 64742-01-4 / RRN 01-2119488707-21, 64742-44-5 / RRN
 01-2119985177-24, 64742-45-6, 64742-52-5 / RRN 01-2119467170-45, 64742-53-6 / RRN
 01-2119480375-34, 64742-54-7 / RRN 01-2119484627-25, 64742-55-8 / RRN
 01-2119487077-29, 64742-56-9 / RRN 01-2119480132-48, 64742-57-0 / RRN
 01-2119489287-22, 64742-58-1, 64742-62-7 / RRN 01-2119480472-38, 64742-63-8,
 64742-65-0 / RRN 01-2119471299-27, 64742-70-7 / RRN 01-2119487080-42, 72623-85-9 /
 RRN 01-2119555262-43, 72623-86-0 / RRN 01-2119474878-16, 72623-87-1 / RRN
 01-2119474889-13

Procedure used to derive the classification according to Regulation (EC) No. 1272/2008 [CLP/GHS]

Classification	Justification
Skin Irrit. 2, H315 Eye Irrit. 2, H319 Aquatic Chronic 3, H412	Calculation method Calculation method Calculation method

Full text of abbreviated H statements

H301 Toxic if swallowed.
 H304 May be fatal if swallowed and enters airways.
 H311 Toxic in contact with skin.
 H314 Causes severe skin burns and eye damage.
 H318 Causes serious eye damage.
 H400 Very toxic to aquatic life.
 H410 Very toxic to aquatic life with long lasting effects.

Full text of classifications [CLP/GHS]

Acute Tox. 3 ACUTE TOXICITY - Category 3
 Aquatic Acute 1 SHORT-TERM (ACUTE) AQUATIC HAZARD - Category 1
 Aquatic Chronic 1 LONG-TERM (CHRONIC) AQUATIC HAZARD - Category 1
 Asp. Tox. 1 ASPIRATION HAZARD - Category 1
 Eye Dam. 1 SERIOUS EYE DAMAGE/EYE IRRITATION - Category 1
 Skin Corr. 1B SKIN CORROSION/IRRITATION - Category 1B

History**Date of issue/ Date of revision**

08/01/2026.

Date of previous issue

01/11/12

Prepared by

Product Stewardship

Indicates information that has changed from previously issued version.

Notice to reader

All reasonably practicable steps have been taken to ensure this data sheet and the health, safety and environmental information contained in it is accurate as of the date specified below. No warranty or representation, express or implied is made as to the accuracy or completeness of the data and information in this data sheet.

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Annex to the extended Safety Data Sheet (eSDS)

Industrial

Identification of the substance or mixture

Product definition	Mixture
Code	467096-DE04
Product name	Rustilo HL 01

Section 1: Title

Short title of the exposure scenario	Use of lubricants and greases in open systems - Industrial
List of use descriptors	Identified use name: Use of lubricants and greases in open systems-Industrial Process Category: PROC01, PROC02, PROC07, PROC08b, PROC09, PROC10, PROC13 Sector of end use: SU03 Subsequent service life relevant for that use: No. Environmental Release Category: ERC04 Specific Environmental Release Category: ATIEL-ATC SPERC 4.Ci.v1

Processes and activities covered by the exposure scenario	Covers use of lubricants and greases in open systems, including application of lubricant to work pieces or equipment by dipping, brushing or spraying (without exposure to heat), e.g. mould releases, corrosion protection, slideways. Includes associated product storage, material transfers, sampling and maintenance activities.
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Section 2 Operational conditions and risk management measures

Section 2.1 Control of worker exposure

Product characteristics:

Physical state:	Liquid, vapour pressure < 0.5 kPa
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Concentration of substance in product:	Covers use of substance/product up to 100 % (unless stated differently)
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Frequency and duration of use:	Covers daily exposures up to 8 hours
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Other operational conditions affecting worker exposure:	Assumes use at not more than 20°C above ambient temperature, unless stated differently. Assumes a good basic standard of occupational hygiene is implemented
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Contributing scenarios: Operational conditions and risk management measures

General measures applicable to all activities:

Avoid direct skin contact with product. Identify potential areas for indirect skin contact. Wear gloves (tested to EN 374) if hand contact with substance likely. Clean up contamination/spills as soon as they occur. Wash off any skin contamination immediately. Provide basic employee training to prevent/minimise exposures and to report any skin problems that may develop. Other skin protection measures such as impervious suits and face shields may be required during high dispersion activities which are likely to lead to substantial aerosol release, e.g. spraying.

Material transfers Manual:

Avoid carrying out activities involving exposure for more than 1 hour per day.

Material transfers Automated process with (semi) closed systems:

Ensure material transfers are under containment or extract ventilation.

Roller, spreader, flow application:

Provide extract ventilation to points where emissions occur.

Spraying:

Carry out in a vented booth or extracted enclosure.

Treatment by dipping and pouring:

Provide a good standard of controlled ventilation (10 to 15 air changes per hour). Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Equipment cleaning and maintenance:

Drain down system prior to equipment break-in or maintenance. Provide a good standard of general ventilation (not less than 3 to 5 air changes per hour). Wear chemically resistant gloves (tested to EN374) in combination with specific activity training. Retain drain-downs in sealed storage pending disposal or for subsequent recycle.

Rustilo HL 01

Use of lubricants and greases in open systems - Industrial

Storage:
Store substance within a closed system.

Section 2.2: Control of environmental exposure

Amounts used:

EU tonnage of risk determining substance per year: 3.81E+01 Tonnes/year

Frequency and duration of use:

Emission days 300

Environment factors not influenced by risk management:

Local freshwater dilution factor 10

Local marine water dilution factor 100

Other operational conditions of use affecting environmental exposure: Negligible wastewater emissions as process operates without water contact.

Release fraction to air (after typical onsite RMMs) 5.00E-05

Release fraction to soil from process (after typical onsite RMMs) 0

Release fraction to wastewater from process (after typical onsite RMMs and before sewage treatment plan) 2E-11

Technical conditions and measures at process level (source) to prevent release: Common practices vary across sites thus conservative process release estimates used.

Technical on-site conditions and measures to reduce or limit discharges, air emissions and releases to soil: Prevent discharge of undissolved substance to or recover from onsite wastewater.

User sites are assumed to be provided with oil/water separators and for waste water to be discharged via public sewer system.

Organisational measures to prevent/limit release from site: Do not apply industrial sludge to natural soils.
Sludge should be incinerated, contained or reclaimed.

Conditions and measures related to municipal sewage treatment plant:

Estimated substance removal from wastewater via on-site sewage treatment 69

Assumed domestic sewage treatment plant flow rate (m³/d) 2.00E+3

Maximum allowable site tonnage (M_{Safe}) based on release following total wastewater treatment removal as product: 726

Conditions and measures related to external treatment of waste for disposal: External treatment and disposal of waste should comply with applicable local and/or national regulations.

Conditions and measures related to external recovery of waste: External recovery and recycling of waste should comply with applicable local and/or national regulations.

Section 3: Exposure estimation and reference to its source

Exposure estimation and reference to its source - Environment

Exposure assessment (environment): Used ECETOC TRA model (May 2010 release).

Exposure estimation and reference to its source - Workers

Exposure assessment (human): The ECETOC TRA tool has been used to estimate workplace exposures unless otherwise indicated.

Section 4: Guidance to check compliance with the exposure scenario

Environment

Guidance is based on assumed operating conditions which may not be applicable to all sites; thus, scaling may be necessary to define appropriate site-specific risk management measures. Further details on scaling and control technologies are provided in SPERC factsheet. If scaling reveals a condition of unsafe use (i.e., RCRs > 1), additional RMMs or a site-specific chemical safety assessment is required. For further information see www.ATIEL.org/REACH_GES

Health

Where other risk management measures/operational conditions are adopted, then users should ensure that risks are managed to at least equivalent levels.

Annex to the extended Safety Data Sheet (eSDS)

Professional

Identification of the substance or mixture

Product definition	Mixture
Code	467096-DE04
Product name	Rustilo HL 01

Section 1: Title

Short title of the exposure scenario	Use of lubricants and greases in open systems - Professional
List of use descriptors	Identified use name: Use of lubricants and greases in open systems-Professional Process Category: PROC01, PROC02, PROC08a, PROC10, PROC11, PROC13 Sector of end use: SU22 Subsequent service life relevant for that use: No. Environmental Release Category: ERC08a, ERC08d Specific Environmental Release Category: ATIEL-ATC SPERC 8.Cp.v1

Processes and activities covered by the exposure scenario	Covers use of lubricants and greases in open systems, including application of lubricant to work pieces or equipment by dipping, brushing or spraying (without exposure to heat), e.g. mould releases, corrosion protection, slideways. Includes associated product storage, material transfers, sampling and maintenance activities.
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Section 2 Operational conditions and risk management measures

Section 2.1 Control of worker exposure

Product characteristics:

Physical state:	Liquid, vapour pressure < 0.5 kPa
Concentration of substance in product:	Covers use of substance/product up to 100 % (unless stated differently)
Frequency and duration of use:	Covers daily exposures up to 8 hours
Other operational conditions affecting worker exposure:	Assumes use at not more than 20°C above ambient temperature, unless stated differently. Assumes a good basic standard of occupational hygiene is implemented

Contributing scenarios: Operational conditions and risk management measures

General measures applicable to all activities:

Avoid direct skin contact with product. Identify potential areas for indirect skin contact. Wear gloves (tested to EN 374) if hand contact with substance likely. Clean up contamination/spills as soon as they occur. Wash off any skin contamination immediately. Provide basic employee training to prevent/minimise exposures and to report any skin problems that may develop. Use suitable eye protection. Avoid direct eye contact with product also via contamination on hands.

Material transfers Manual:

Avoid carrying out activities involving exposure for more than 1 hour per day.

Roller, spreader, flow application:

Natural ventilation is from doors, windows etc. Controlled ventilation means air is supplied or removed by a powered fan. Avoid carrying out activities involving exposure for more than 4 hours. Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Spraying:

Natural ventilation is from doors, windows etc. Controlled ventilation means air is supplied or removed by a powered fan. Avoid carrying out activities involving exposure for more than 1 hour per day. Wear a respirator conforming to EN140 with type A/P2 filter or better. Wear suitable coveralls to prevent exposure to the skin. Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Treatment by dipping and pouring:

Natural ventilation is from doors, windows etc. Controlled ventilation means air is supplied or removed by a powered fan.

Equipment cleaning and maintenance:

Drain down system prior to equipment break-in or maintenance. Natural ventilation is from doors, windows etc. Controlled ventilation means air is supplied or removed by a powered fan. Avoid carrying out activities involving

Rustilo HL 01

Use of lubricants and greases in open systems - Professional

exposure for more than 4 hours. Retain drain-downs in sealed storage pending disposal or for subsequent recycle.

Storage:
Store substance within a closed system.

Section 2.2: Control of environmental exposure

Amounts used:

EU tonnage of risk determining substance per year: 2.24E+01 Tonnes/year

Frequency and duration of use:

Emission days 365

Environment factors not influenced by risk management:

Local freshwater dilution factor 10

Local marine water dilution factor 100

Other operational conditions of use affecting environmental exposure: Negligible wastewater emissions as process operates without water contact.

Release fraction to air (after typical onsite RMMs) 1.00E-04

Release fraction to soil from process (after typical onsite RMMs) 1E-03

Release fraction to wastewater from process (after typical onsite RMMs and before sewage treatment plan) 5.00E-04

Technical conditions and measures at process level (source) to prevent release: Common practices vary across sites thus conservative process release estimates used.

Technical on-site conditions and measures to reduce or limit discharges, air emissions and releases to soil: Prevent discharge of undissolved substance to or recover from onsite wastewater.

Organisational measures to prevent/limit release from site: Do not apply industrial sludge to natural soils. Sludge should be incinerated, contained or reclaimed.

Conditions and measures related to municipal sewage treatment plant:

Estimated substance removal from wastewater via on-site sewage treatment 69

Assumed domestic sewage treatment plant flow rate (m³/d) 2.00E+3

Maximum allowable site tonnage (M_{Safe}) based on release following total wastewater treatment removal as product: 3.3

Conditions and measures related to external treatment of waste for disposal: External treatment and disposal of waste should comply with applicable local and/or national regulations.

Conditions and measures related to external recovery of waste: External recovery and recycling of waste should comply with applicable local and/or national regulations.

Section 3: Exposure estimation and reference to its source

Exposure estimation and reference to its source - Environment

Exposure assessment (environment): Used ECETOC TRA model (May 2010 release).

Exposure estimation and reference to its source - Workers

Exposure assessment (human): The ECETOC TRA tool has been used to estimate workplace exposures unless otherwise indicated.

Section 4: Guidance to check compliance with the exposure scenario

Environment

Guidance is based on assumed operating conditions which may not be applicable to all sites; thus, scaling may be necessary to define appropriate site-specific risk management measures. Further details on scaling and control technologies are provided in SPERC factsheet. If scaling reveals a condition of unsafe use (i.e., RCRs > 1), additional RMMs or a site-specific chemical safety assessment is required. For further information see www.ATIEL.org/REACH_GES

Health

Where other risk management measures/operational conditions are adopted, then users should ensure that risks are managed to at least equivalent levels.